Formula sheet - Probability & Statistics (Premaster MTPS)

2026 – 2027

Probability Rules:

Complement rule:	$P(A') = 1 - P(A)$
General addition rule:	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional probability:	$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
Law of total probability:	Assume S can be partitioned into B_1, B_2, …, B_k: $P(A) = \sum_{i=1}^{k} P(A \mid B_i) \cdot P(B_i)$
Independence:	for independent events A and B: $P(A \cap B) = P(A) \cdot P(B)$
Bayes' theorem:	$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$

Discrete and continuous random variables:

	Discrete random variables:	**Continuous random variables:**
Sample space:	Individual, separate values e.g., $0, 1, 2, \ldots$	Spectrum of values, e.g., the interval $[0, 1]$.
Application:	Counting / categorizing	Measurements
Probability concept:	Probability mass function $f(x) = P(X = x)$ • $f(x) \geq 0$ for all x • $\sum_x f(x) = 1$	Probability density function $f(x)$ • $f(x) \geq 0$ for all x • $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$
CDF	$F(x) = P(X \leq x) = \sum_{y:y \leq x} f(y)$	$F(x) = P(X \leq x) = \int_{y=-\infty}^{x} f(y)\,\mathrm{d}y$
Expected value:	$E(X) = \sum_x x \cdot f(x)$	$E(X) = \int_{x=-\infty}^{\infty} x \cdot f(x)\,\mathrm{d}x$
Variance:	$V(X) = \sum_x (x - E(X))^2 \cdot f(x)$ $V(X) = E(X^2) - [E(X)]^2$	$V(X) = \int_{x=-\infty}^{\infty} (x - E(X))^2 \cdot f(x)\,\mathrm{d}x$ $V(X) = E(X^2) - [E(X)]^2$
Standard deviation:	$\sigma(X) = \sqrt{V(X)}$	$\sigma(X) = \sqrt{V(X)}$

Common probability distributions:

Distribution	PMF/PDF	CDF	Mean	Variance
		DISCRETE DISTRIBUTIONS		
Uniform(a,b) (values $x_1,\dots,x_n$ step size $d=\frac{b-a}{n-1}$)	$f(x)=\frac{1}{n}$ $(x=x_1,\dots,x_n)$	$F(x)=\begin{cases}0, & x<a,\\ \frac{k}{n}, & x_k\le x<x_{k+1},\\ 1, & x\ge b.\end{cases}$	$\frac{a+b}{2}$	$d^2\cdot\frac{n^2-1}{12}$
Bernoulli(p)	$f(x)=\begin{cases}1-p, & x=0,\\ p, & x=1.\end{cases}$	$F(x)=\begin{cases}0, & x<0,\\ 1-p, & 0\le x<1,\\ 1, & x\ge 1.\end{cases}$	p	$p\cdot(1-p)$
Binomial(n,p)	$f(x)=\binom{n}{x}\cdot p^x\cdot(1-p)^{n-x}$ $(x=0,1,\dots,n)$	$F(x)=\sum_{i=0}^{x}\binom{n}{i}\cdot p^i\cdot(1-p)^{n-i}$	$n\cdot p$	$n\cdot p\cdot(1-p)$
Geometric(p)	$f(x)=p\cdot(1-p)^{x-1}$ $(x=1,2,\dots)$	$F(x)=\sum_{i=1}^{x}p\cdot(1-p)^{i-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Pois(λ)	$f(x)=e^{-\lambda}\cdot\frac{\lambda^x}{x!}$ $(x=0,1,\dots)$	$F(x)=\sum_{i=0}^{x}e^{-\lambda}\cdot\frac{\lambda^i}{i!}$	λ	λ
		CONTINUOUS DISTRIBUTIONS		
Uniform(a,b)	$f(x)=\begin{cases}\frac{1}{b-a}, & a\le x\le b,\\ 0, & \text{otherwise.}\end{cases}$	$F(x)=\begin{cases}0, & x<a,\\ \frac{x-a}{b-a}, & a\le x<b,\\ 1, & x\ge b.\end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$N(\mu,\sigma)$	$f(x)=\frac{1}{\sigma\cdot\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$F(x)=\Phi\left(\frac{x-\mu}{\sigma}\right)$	μ	σ^2
Exponential(λ)	$f(x)=\begin{cases}\lambda e^{-\lambda x}, & x\ge 0,\\ 0, & \text{otherwise.}\end{cases}$	$F(x)=\begin{cases}1-e^{-\lambda x}, & x\ge 0\\ 0, & x<0\end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Joint probability distributions:

Marginal distributions:	$f_X(x)=\sum_y f_{X,Y}(x,y)$ (discrete) $f_X(x)=\int_y f_{X,Y}(x,y)\,\mathrm{d}y$ (continuous)
Conditional probability:	$f_{X\mid Y}(x\mid y)=\frac{f_{X,Y}(x,y)}{f_Y(y)}$ (discrete)
Covariance:	$\mathrm{Cov}(X,Y)=E\left[(X-E(X))\cdot(Y-E(Y))\right]=E(XY)-E(X)\cdot E(Y)$
Correlation coefficient:	$\rho(X,Y)=\frac{\mathrm{Cov}(X,Y)}{\sigma(X)\cdot\sigma(Y)}$

Functions of random variables:

Linearity of expectation:	$E(aX+bY+c)=a\cdot E(X)+b\cdot E(Y)+c$
Variance of functions of random variables:	$V(aX+bY)=a^2\cdot V(X)+b^2\cdot V(Y)+2a\cdot b\cdot\mathrm{Cov}(X,Y)$

Central limit theorem:

Let $X_1,X_2,\dots,X_n$ be independent and identically distributed random variables with mean μ and standard deviation σ. Then the following holds:

- the sum $\sum_{i=1}^{n} X_i = X_1 + X_2 + \ldots + X_n \approx N(n \cdot \mu, \sqrt{n} \cdot \sigma)$.
- the average $\overline{X} = \frac{X_1 + X_2 + \ldots + X_n}{n} \approx N(\mu, \sigma/\sqrt{n})$.

Point estimators:

Let $\hat{\Theta}$ be an estimator (random variable) and $\hat{\theta}$ an estimate (outcome) of a population parameter θ.

Sample mean:	$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n}$
Sample variance:	$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$
Bias:	$\text{Bias}(\hat{\Theta}) = E(\hat{\Theta}) - \theta$
Standard error:	$\text{SE}(\hat{\Theta}) = \sqrt{V(\hat{\Theta})}$
Mean squared error:	$\text{MSE}(\hat{\Theta}) = E\left((\hat{\Theta} - \theta)^2\right) = V(\hat{\Theta}) + \text{Bias}(\Theta)^2$
Relative efficiency:	$\frac{\text{MSE}(\hat{\Theta_1})}{\text{MSE}(\hat{\Theta_2})}$
Maximum likelihood:	• Define the likelihood function: $L(\theta) = \prod_{i=1}^{n} f(x_i; \theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \ldots \cdot f(x_n; \theta)$ • If necessary, take the log likelihood function $\ln L(\theta)$ (as products becomes sums when taking logs). • Determine the value of θ that maximizes $L(\theta)$ (or $\ln L(\theta)$).

Interval estimators:

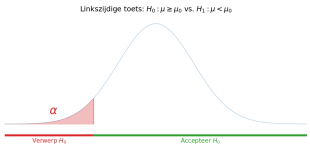
z**-value:**	**TI:** $z_{\alpha/2} = \text{InvNorm}(\text{area} = 1 - \alpha/2, \mu = 0, \sigma = 1)$ **Casio:** $z_{\alpha/2} = \text{InvN}(\text{area} = 1 - \alpha/2, \sigma = 1, \mu = 0)$
t**-value:**	**TI:** $t_{n-1,\alpha/2} = \text{InvT}(\text{area} = 1 - \alpha/2, \text{df} = n - 1)$ **Casio:** $t_{n-1,\alpha/2} = \text{Invt}(\text{area} = \alpha/2, \text{df} = n - 1)$
$100(1-\alpha)\%$**-confidence interval (CI) for the mean** μ **(**σ **known):**	$[\overline{X} - z_{\alpha/2} \cdot \sigma/\sqrt{n}, \quad \overline{X} + z_{\alpha/2} \cdot \sigma/\sqrt{n}]$
$100(1-\alpha)\%$**-confidence interval (CI) for the mean** μ **(**σ **unknown):**	$[\overline{X} - t_{n-1,\alpha/2} \cdot S/\sqrt{n}, \quad \overline{X} + t_{n-1,\alpha/2} \cdot S/\sqrt{n}]$
Minimum sample size for a $100(1-\alpha)\%$**-CI with error margin** $\pm E$ **(**σ **known):**	$n \geq \left(\frac{z_{\alpha/2} \cdot \sigma}{E}\right)^2$
Minimum sample size for a $100(1-\alpha)\%$**-CI with error margin** $\pm E$ **(**σ **unknown):**	• Determine initial value for n using formula above (as if $\sigma = s$). • Fill in $y_1 = s/\sqrt{X} \cdot t_{X-1,\alpha/2}$ (see t-value). • Generate table starting from initial value and determine the smallest n such that $y_1 \leq E$.

Hypothesis testing

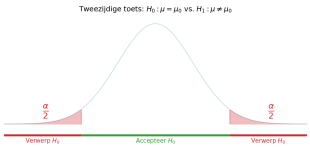
Test name	Test statistic	Probability distribution (under H_0)
Tests for the mean μ of a normal distribution		
σ known	$Z = \frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}}$	$N(0,1)$
σ unknown	$T = \frac{\overline{X}-\mu_0}{S/\sqrt{n}}$	$t(\text{df} = n-1)$
Chi-square test (χ^2)		
Chi-square goodness-of-fit	$X^2 = \sum_i \frac{(O_i-E_i)^2}{E_i}$	χ^2 (df = #categories-1)
Correlation test ($\rho = 0$)		
Significant correlation	$T = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}}$	$t(\text{df} = n-2)$

One-sided and two-sided hypothesis tests

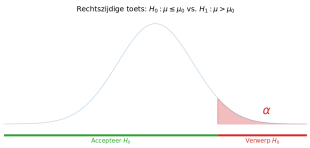
Left-sided
$H_0 : \mu \geq \mu_0$ vs. $H_1 : \mu < \mu_0$
Acceptance region: $[g, \infty)$
Critical region: $(-\infty, g)$



Two-sided
$H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$
Acceptance region: $[g_1, g_2]$
Critical region: $(-\infty, g_1)$ and (g_2, ∞)



Right-sided
$H_0 : \mu \leq \mu_0$ vs. $H_1 : \mu > \mu_0$
Acceptance region: $(-\infty, g]$
Critical region: (g, ∞)



Critical regions:

Probability distribution (under H_0)	**Left-sided** $(-\infty, g)$	**Two-sided** $(-\infty, g_1)$ and (g_2, ∞)	**Right-sided** (g, ∞)
$N(0,1)$	InvNorm with area $= \alpha$	InvNorm with area $= \alpha/2$ and $1-\alpha/2$	InvNorm with area $= 1-\alpha$
$t(\text{df})$	InvT with area $= \alpha$	InvT with area $= \alpha/2$ and $1-\alpha/2$	InvT with area $= 1-\alpha$
Critical region that needs to be solved with the "numeric solver" or "SolveN":			
$\chi^2(\text{df})$	-	-	**TI:** $\chi^2\text{cdf}(g, 10^{99}, \text{df}) = \alpha$ or **Casio:** SolveN(ChiCD$(x, 10^{99}, \text{df}) = \alpha, x)$

N.B.: when using Casio: (i) the order is σ, μ, and (ii) Invt *computes boundary values with area on the right, so replace area by $1 -$ area where applicable.*

p-values

Probability distribution (under H_0)	**Test statistic**	**Left-sided**	**Two-sided**	**Right-sided**
$N(0,1)$	Z	$p = P(Z \leq z)$	$p =$ minimum of $P(Z \leq z)$ and $P(Z \leq z)$	$p = P(Z \geq z)$
t(df)	T	$p = P(T \leq t)$	$p =$ minimum of $P(T \leq t)$ and $P(T \leq t)$	$p = P(T \geq t)$
χ^2(df)	X^2	$p = P(X^2 \leq \chi^2)$	$p =$ minimum of $P(X^2 \leq \chi^2)$ and $P(X^2 \leq \chi^2)$	$p = P(X^2 \geq \chi^2)$
		Compare with α	Compare with $\alpha/2$	Compare with α

Type-II errors

Type-II error β for a z-test (given $H_1 : \mu = \mu_1$):

- Transform critical region for $N(0,1)$ to a critical region for $N(\mu_0, \sigma/\sqrt{n})$ using $x \mapsto \mu_0 + x \cdot \sigma/\sqrt{n}$.
- $\beta = 1 - P(\overline{X}$ in critical region $|\mu = \mu_1)$
  $= 1 - \text{normalcdf}(\ldots, \ldots, \mu_1, \sigma/\sqrt{n})$

Minimum sample size n such that $\beta \leq \beta_0$ and $\mu_1 = \mu_0 + \delta$: $n \geq \left(\frac{(z_{\alpha/2} + z_{\beta_0}) \cdot \sigma}{\delta}\right)^2$

Linear regression and correlation

Simple linear regression model: $Y = \beta_0 + \beta_1 \cdot x + \epsilon$

Least squares coefficients: $\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n \cdot \overline{x} \cdot \overline{y}}{\sum_{i=1}^n x_i^2 - n \cdot \overline{x}^2} = \frac{\overline{xy} - \overline{x} \cdot \overline{y}}{\overline{x^2} - \overline{x}^2}$
$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \cdot \overline{x}$

Variance of error terms: $\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{\sum_{i=1}^n y_i^2 - \hat{\beta}_0 \cdot \sum_{i=1}^n y_i - \hat{\beta}_1 \cdot \sum_{i=1}^n x_i y_i}{n-2}$

Pearson's correlation coefficient: $r(x,y) = \frac{\text{Cov}(x,y)}{\sqrt{s_X^2 \cdot s_Y^2}} = \frac{\overline{xy} - \overline{x} \cdot \overline{y}}{\sqrt{(\overline{x^2} - \overline{x}^2) \cdot (\overline{y^2} - \overline{y}^2)}}$